

MANAGEMENT OF ACUTE GOUT AND CHRONIC GOUT IN PRIMARY CARE

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KEY TAKE-HOME MESSAGES

- Gout is a chronic condition and appropriate long-term ongoing treatment should be instituted if indicated
- Provide lifestyle management to all patients with gout
- Do not delay ULT for patients with gout who meet ULT treatment criteria* (Table 1)
- “Start low, go slow” with ULT
- Treatment targets for patients on ULT
 - < **360 µmol/L** for non-tophaceous gout
 - < **300 µmol/L** for tophaceous gout
- For patients on ULT, do not stop ULT during an acute flare
- ULT can be initiated during acute flare. There is no need to wait until the acute flare has resolved if adequate treatment is provided for the acute flare.
- Monitor serum uric acid in gout patients at least every 6 months (more frequently as clinically indicated for patients on active titration of ULT)

***Table 1 ULT Treatment Criteria**

- | |
|--|
| • Frequent acute gout flares (≥ 2 per year) |
| • Tophaceous Gout |
| • Clinical or radiological findings of gouty arthropathy |
| • History of urolithiasis (urate calculi) |
| • Urate nephropathy or any renal insufficiency (CKD 3 and above) |

HISTORY AND PRESENTATION

- Acute onset of severe pain (reaching maximum pain within 6 - 12 hours, up to 24 hours) and asymmetrical swelling affecting 1st metatarsophalangeal joint (MTPJ) of the big toes, ankle joints, and occasionally knees and small joints of the hands such as PIP joints
- Erythema of the affected joint(s) with warmth
- Recurrent 1st MTP joint symptoms (most common)
- Strong family history of gout
- Remission of symptoms within a few days to 1 week even without active treatment
- Beware of “grumbling gout” which is an uncontrolled and untreated gout attack
- Men > women (post-menopausal)
- May present with symmetrical polyarthritis and fever – infection/ septic arthritis will need to be excluded

EXCLUDE OTHER CO-MORBIDITIES / RISK FACTORS

- Diabetes Mellitus
- Hypertension
- Hyperlipidemia
- Chronic kidney disease
- Excessive alcohol intake
- Excessive intake of purine-rich food from animal sources (animal organs, red meat *etc.*)
- Excessive intake of fructose (corn syrup, sweetened beverages)
- Smoking
- Obesity
- Usage of diuretic (diuretic increases uric acid level by directly increasing uric acid absorption and decreasing uric acid secretion. Diuretic also can indirectly cause volume depletion which plays an important role in diuretic-induced hyperuricemia. It is dose-dependent but prophylactic treatment with allopurinol is not necessary unless symptomatic), Cyclosporine, and other drugs

PHYSICAL EXAMINATION AND SIGNS

- Body temperature
- Obesity (BMI)
- Blood pressure
- Tender and swollen joints (monoarticular or asymmetric polyarticular) - commonly 1st MTPJ of the foot (Podagra), small joint of fingers (PIPJs), tarsal joint of the foot, ankle, elbow and occasionally knee
- Deformity of affected joint(s) in chronic gout
- Abdominal exam if history suggestive of alcoholic liver disease, PUD. Urate nephropathy seldom causes obstructed / bilateral ballotable kidneys

Podagra (Swollen, erythematous and tender 1 st MTPJ)	Tophus(i)
 Podagra	

INVESTIGATIONS

- The gold standard for diagnosis: Presence of characteristic urate crystal (negatively birefringent needle-shaped) in joint aspiration fluid of affected joint or presence of a tophus proven to contain characteristic urate crystals by chemical means or polarized light microscopy (not done in the polyclinic setting)
- Baseline investigations* (referenced from [Recommended Care Components for Gout](#)):
 - **FBC** - to exclude infection, haematological disorder and anaemia associated with prolonged NSAIDs use. However, be aware that during acute gout attacks, you can get high TW and polymorphs. Consider repeating 2 - 8 weeks after initiation of allopurinol
 - **Creatinine** - to exclude renal disease leading to hyperuricaemia and renal disease secondary to urate nephropathy
 - **Uric Acid** – as a baseline and as a treatment target. Serum uric acid should not be used to confirm or exclude gout
 - Recommended to monitor serum uric acid minimally every 6 months in gout patients
 - **HYPERURICEMIA ≠ GOUT**
 - **ALT and AST** – as a baseline to exclude hepatic impairment and for follow up 2 - 8 weeks after initiation of allopurinol. Fatty liver is also commonly noted in patients with metabolic syndrome
 - **Plasma Glucose** – to detect the presence of diabetes mellitus, insulin resistance (features of metabolic syndrome)
 - **Fasting Lipids** – to look out for high triglyceride and low HDL cholesterol (features of metabolic syndrome)
 - **ESR** – can exclude other suspected inflammatory arthritis if history is not totally consistent with gout. However, may not be helpful as often it will be high in the presence of any acute red, hot swollen joint. (not routinely done if the diagnosis is very suggestive of gout)
 - ESR being an acute phase reactant and non-specific inflammatory marker, should always be interpreted within specific clinical context – including patients' age and gender
 - A quick acceptable formula to correct reference range for ESR will be:
 - **Male:** Age in years/2
 - **Female:** (Age in years + 10)/2
 - E.g. for a female patient of 60 years old, an ESR level of 35 [(60 + 10)/2] is a reasonable threshold to be considered clinically significant
 - In the event that ESR is abnormal (after adjusting for age and gender) patient should be reviewed for possible etiologies (e.g. systemic or localised inflammation or infectious diseases, tissue injury, trauma, smoking etc.)

- If there is no clear identifiable etiology and in the absence of red flags (e.g. fever, severe infections, weight loss, inflammatory joint pain etc) – a raised ESR can be trended
- If ESR is persistently raised (without specific symptoms), a referral to internal medicine can be considered.
- **CK** – can consider at baseline or if there is concern of colchicine related myopathy
- **X-Ray of Relevant Joints** - to look out for periarticular erosions seen in chronic gout, especially where it is difficult to differentiate history from other inflammatory arthritides or in the presence of significant joint deformity that results in impairment of function (not routinely done if not indicated)



Features of Gouty Arthritis on Plain X-Ray

- Soft tissue swelling and edema (in acute presentations)
- Periarticular or juxta-articular erosions - classically termed as “punched-out” lesions (see picture)
- Interosseous tophi, joint space narrowing with sclerosis, deformity and subluxation (in late chronic tophaceous gout)

GOUT DIAGNOSIS

- 1st episode of acute gout
- 1st episode of acute onset of periarticular gout
 - Periarticular gout can present at bursae (knee & olecranon bursae) and tendon sheaths such as the arch of feet or at Achilles tendon area. Consider gout as a diagnosis for recurrent painful soft tissue swelling at the Achilles tendon insertion at the calcaneum
- Frequent gout flares (≥ 2 episodes per year)
- Chronic tophaceous gout
- Inter-critical gout:
 - Inter-critical gout is the time between gout attacks. (e.g. Patients might have a gout attack 3 months ago and by the time they come for consults, it has resolved)
 - This period of gout free attack is considered inter-critical
 - It is important to recognise this and start appropriate ULT if indicated. If the gout is left untreated, the time between attacks may be shortened and the symptoms can become more severe and prolonged

DIFFERENTIAL DIAGNOSES

- Septic arthritis (May be difficult to distinguish from acute gout flares, can co-exist with acute gout flares on rare occasions)
- Trauma (trauma itself can trigger acute gout flare)
- Pseudogout (calcium pyrophosphate crystal deposition)
 - More commonly affect knees, wrists, hips but can occur at any joint
- Cellulitis (extension of inflammatory signs and symptoms to the non-articular area accompanied by systemic signs such as fever / chills)
- To consider pes planus with hallux valgus as causes of the foot pain that the patient claimed to be gout, especially in female patients

ASYMPTOMATIC HYPERURICEMIA

- Gout should not be diagnosed based solely on elevated serum uric acid
- Avoid performing routine / screening serum uric acid levels in patients without history or clinical findings suggestive of gout
- Non-pharmacological advice similar to those given to gout patients can be provided as part of the management approach to asymptomatic hyperuricemia
- Usually, no further workup or pharmacological treatment is required

TREATMENT GOALS

- Lower serum uric acid level, decrease acute flare frequency, promote crystal dissolution and prevent crystal / tophi formation
- Maintain serum uric acid below **< 360 µmol/L** in non-tophaceous gout
- In tophaceous gout, maintain serum uric acid below **< 300 µmol/L**

Non-Pharmacological

- Low purine diet (refer to a dietician)
- Lifestyle modification (reduce weight, avoid smoking and alcohol intake)
- Adequate fluid intake (if not contraindicated)
- Beware of iatrogenic causes such as usage of diuretics. Review medication list and consider reducing diuretics e.g. in stable CHF or hypertensive patients as first-line management

Pharmacological

Acute Management:

- **Colchicine:** 500 mcg BD / TDS (3 - 5 days) **± NSAIDs**. To advise on side effects such as gastritis and diarrhoea
 - Types of NSAIDs: Indomethacin, Naproxen, Ibuprofen and Diclofenac Sodium *etc.*
 - NSAIDs should be used with caution or avoided in patient with renal impairment (CKD 3 and above)
 - NSAIDs should be avoided in patients taking concomitant antiplatelet / anticoagulant

- NSAIDs / selective cyclo-oxygenase inhibitors should be used with extreme caution (consider using a lower dose) in the elderly even if serum creatinine is within the normal reference interval
- Exclude contraindications to the initiation and prolonged usage of NSAIDs: poorly controlled asthma or aspirin intolerant asthma, uncontrolled hypertension and established cardiovascular disease. Consider prescribing prophylactic PPI to avoid gastric side effects
- NSAIDs should be prescribed at the lowest dose or shortest duration possible
 - Avoid prescribing NSAIDs for more than 2 weeks of continuous usage
- Colchicine dose needs to be reduced in renal impairment and avoided in moderate/severe renal impairment
- 0.5 mg twice or three times daily during acute attacks for 3 - 5 days
- If eGFR $\leq$ 30ml/min and in severe hepatic impairment, 0.5 mg 2 - 3 times/week
- Beware of increased risk of myopathy with concomitant use with statins (less likely with fluvastatin and rosuvastatin) that are also metabolized by the CYP3A4 pathway
 - When patients are given colchicine, consider temporarily withhold statins if appropriate to reduce the risk of adverse drug reactions
- **Prednisolone** $\leq$ 30 mg/day (0.5 mg/kg or whichever is lower) in divided doses $\pm$ colchicine 500 mcg TDS (3 - 5 days)
 - It is recommended to use short courses of glucocorticoids in elderly people and those with renal insufficiency, hepatic dysfunction, cardiac failure, peptic ulcer disease and hypersensitivity to NSAIDs
- **Joint aspiration and intra-articular injection** with triamcinolone (Beware of septic joints, to be performed only by qualified doctors) of the affected joint, e.g. knee
- **Other alternative analgesia** includes: Anarex, Panadeine, Tramadol and Paracetamol
- If the patient has been on allopurinol, continue the medication during the acute attack
 - **DO NOT STOP ALLOPURINOL DURING ACUTE FLARE OF GOUT**
- It is common for patients who are started or on increasing doses of allopurinol to have breakthrough attacks. It is important not to stop or lower the dose of allopurinol during this period of attacks. Colchicine 500 mcg OM / BD should be used as prophylaxis whenever initiate or adjusting allopurinol dose until target serum uric acid is achieved (up to 6 months)
- If an acute flare occurs during prophylaxis: use the higher acute flare treatment dose and reinstate the prophylactic dose once the acute flare resolves

MAINTENANCE SERUM URATE LOWERING THERAPY (ULT)

Indications

- Frequent acute gout flares (significant in frequency and severity: ≥ 2 attacks per year)
 - Lowering threshold for ULT can be considered even if attacks are infrequent but are prolonged or interfere with activities.
(Discuss with the patients for informed decision)
- Tophaceous gout
- Clinical or radiological changes of gout i.e. erosions suggestion signs of chronic gouty arthropathy
- Urate nephropathy / Urate calculi
- Consider treating gout in patients with any significant renal insufficiency (CKD 3 and above)
- Other clinical scenarios where starting ULT can be considered and discussed with patients base on their individualised risks:
 - 1st gout flare in patients with high serum uric acid level (> 535 $\mu\text{mol/L}$)
 - 1st gout flare in patients with history of urolithiasis

Xanthine Oxidase Inhibitor

- Allopurinol
 - Start 50 - 100 mg/day, increase every 2 - 8 weeks using serum uric acid and clinical features as a guide till target serum uric acid achieved
 - Max dose is 800 - 900 mg a day for patients with normal renal function
 - **SCAR: Severe Cutaneous Adverse Reaction**
 - Allopurinol induced SCARs are uncommon (about 3 in 1000) and include: SJS, TEN, DRESS
 - Most SCAR occurs within weeks to months of therapy initiation
 - Risk factors for SCAR: **RASHES**
 - Renal impairment
 - Agent (concomitant use: e.g. diuretics)
 - Starting dose: high starting dose
 - HLA-B*5801
 - Escalation: rapid increase in dose
 - Seniority: elderly patients
 - Monitor closely for SCAR in initial weeks to months after treatment initiation (document on advising patients of early signs of SCAR: fever, rashes, oral ulcers / severe sore throat, red or sore eyes)
 - FBC, ALT, AST should be repeated 2 - 8 weeks after starting / escalation of allopurinol to look for cytopenias / eosinophilia, deranged LFTs
 - Increase allopurinol dose by 50 - 100 mg each review till the target serum urate level achieved

- Renal adjusting maintenance doses of allopurinol may lead to undertreatment. Therefore, it is more important to reduce the initial starting dose of allopurinol and slowly uptitrate to maintenance dose in CKD patients to reduce risk of adverse side effects

Proposed starting dose of Allopurinol per estimated GFR		
eGFR (ml/min/1.73m ²)	Starting dose of Allopurinol	Remarks
< 5	50 mg/week	CKD 4/5 patients should be considered to be referred to RAI SOC for management
5 - 15	50 mg twice a week	
16 - 30	50 mg EOD	
31 - 45	50 mg OM	
46 - 60	50 – 100 mg OM	
> 60	100 mg OM	

- In patients with CKD, caution is required for doses of allopurinol that is > 300mg daily and may be considered with appropriate patient education and monitoring for potential toxicity (e.g. rash, pruritus, elevated transaminases)
- In patients with severe renal impairment (CrCl < 10mL/min) to consider keeping max daily dose of allopurinol at 100mg

Uricosuric Agents

- Probenecid acid (if no renal impairment and no renal stones)
 - Probenecid is not recommended for patients with CrCl < 50 mL/min (ineffective)
 - Contraindicated in patients with renal stone
 - May induce hemolytic anaemia in patients with G6PD deficiency
 - The initial dose is 250 mg BD for the first two to four weeks followed by 250 mg increments every four to six weeks till a maximum dose of 2 grams per day
 - To be taken in divided doses
 - Advise adequate hydration
 - It decreases the tubular excretion of salicylate and Penicillin
 - It increases the serum frusemide concentration and augments its diuretic action

- Benzbromarone (**Not available in NUP, not commonly used locally**)
 - Start at 25 - 50 mg/day, typical dosage is 50 - 100 mg/day (maximum dose 200 mg/day, but doses > 100 mg/day more likely associated with elevated liver enzymes)
 - It is ineffective when the CrCl < 20 mL/min and can be used in moderate renal impairment. It is not available in primary care.
 - Liver function tests need to be monitored, as it is associated with hepatotoxicity

Non-Purine Xanthine Oxidase Inhibitor

- Febuxostat (**Not available in NUP**)
 - This drug has shown to be more effective in reducing serum uric acid than allopurinol 100 - 300 mg daily. Effect of febuxostat has not been compared with higher doses of allopurinol
 - It is costlier and current usage is restricted to rheumatologists only
 - It is not available in primary care
 - This should be reserved for patients where maximum doses of allopurinol or allopurinol + probenecid are ineffective, or severe allopurinol allergy where desensitisation is contraindicated

Others

- Losartan (modest uricosuric effect)
- Fenofibrate (modest uricosuric effect)
- Atorvastatin

Prophylaxis

- **Colchicine** 0.5 mg once or twice daily should be started at the initiation of urate-lowering therapy until target serum uric acid is achieved (up to 6 months). The dose of colchicine may need modification depending on renal impairment
- If eGFR ≤ 30ml/min and in severe hepatic impairment, 0.5 mg 2 - 3 times/week

CAUTION: Use colchicine with caution in patients who are on concomitant statins, macrolides or other drugs that are metabolized by CYP3A4 pathway as it potentiates the risk of colchicine toxicity which is associated with pancytopenia and myopathy

REFERRAL TO RHEUMATOLOGIST

Patients with RECURRENT / CHRONIC GOUT can be referred to a rheumatologist if they have ANY of the following:

- Patients tested positive on pharmacogenetics to be HLA-B5801 **positive** AND any of the following:
 - has not been started on allopurinol
 - patients prefer alternatives (e.g. febuxostat) when allopurinol has only been given for **less** than 3 months
 - patients prefer alternatives (e.g. febuxostat) when allopurinol has been given for **more** than 3 months but requires further up-titration as gout treatment targets have not been met

Complicated gout despite adherence to urate-lowering therapy (ULT)

- Significant chronic kidney disease (CKD 4/5)
- Tophaceous gout
- Hypersensitivity or serious adverse effects to ULT
- ≥ 4 attacks/year despite reaching target serum uric acid levels
- Serum uric acid not at target despite moderately high dose (500 mg/day of allopurinol) + trigger minimized / removed if appropriate:
e.g. diuretics

HLA-B5801 TESTING

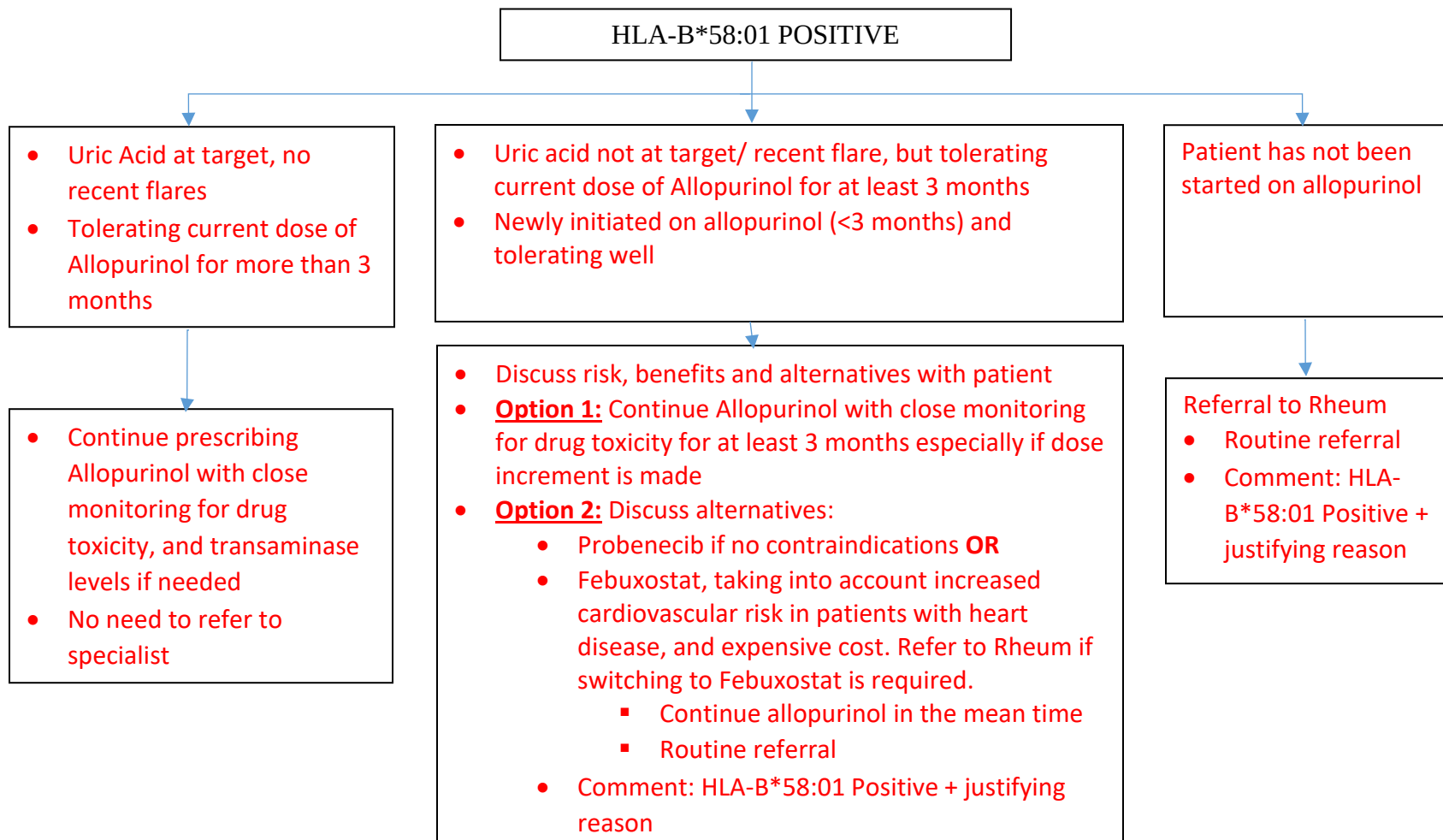
HLA-B5801 testing is to prevent severe cutaneous adverse reactions (SCARs) in allopurinol use

Currently, we do not offer the test as standard screening for patients being started on allopurinol therapy

The overall value of routine genotyping in the Singapore population before initiation of allopurinol is limited

In the event if any patient is tested positive for HLA-B 5801 clinical consideration is reflected below:

HLA-B*58:01 POSITIVE ALLOPURINOL Workflow



RECOMMENDED CARE COMPONENTS FOR GOUT

Recommended Care Component	Minimum Frequency*	Remarks
Serum Uric Acid	At baseline and thereafter as clinically indicated	Frequency of monitoring to be tailored based on clinical indication (suggested every 6 months or more frequent if clinically indicated). Target serum uric acid is below < 360 µmol/L for non-tophaceous gout, and below < 300 µmol/L for tophaceous gout.
Renal Function Monitoring	At baseline	Baseline creatinine and eGFR to exclude renal disease. Consider yearly monitoring if patient on periodic NSAID use. Frequency of monitoring to be tailored based on clinical indication.
Alanine Aminotransferase (ALT), Aspartate Transaminase (AST)	At baseline	Baseline to exclude hepatic impairment and to detect fatty liver (feature of metabolic syndrome). If starting Allopurinol, consider repeating 2 - 8 weeks after initiation and as clinically indicated thereafter to monitor for deranged liver function in severe cutaneous adverse reaction (SCAR).
Full Blood Count (FBC)	At baseline	Consider FBC to exclude infection or haematological disorders. If starting Allopurinol, consider repeating 2 - 8 weeks after initiation and as clinically indicated thereafter to monitor for leukocytosis and eosinophilia in SCAR.
Erythrocyte Sedimentation Rate (ESR)	At baseline (where applicable)	Consider ESR to exclude other suspected inflammatory arthritis.
Creatine Kinase level	At baseline	Consider CK before starting Colchicine to document levels and exclude muscle pathology. Consider repeating 2 - 8 weeks after initiation and 3 - 6 monthly thereafter to monitor for Colchicine myopathy.
Plasma Glucose	At baseline; consider yearly thereafter	To detect insulin resistance and diabetes mellitus (features of metabolic syndrome).
Fasting Lipids	At baseline; consider yearly thereafter	To detect dyslipidaemia (feature of metabolic syndrome). All patients should be risk stratified with targets of treatment tailored accordingly.
X-ray of Relevant Joints	If clinically indicated	Consider imaging to differentiate from other inflammatory arthritides.
Blood Pressure Measurement	Twice a year	To personalise target blood pressure based on patient's risk factors.
Weight and BMI Assessment	Twice a year	Keep BMI < 23kg/m ² (for non-Asian population, keep BMI < 25 kg/m ²).
Assessment of Diet and Lifestyle	Annually	All patients should be advised on low purine diet and lifestyle modification (alcohol avoidance and smoking cessation).

* More frequently if clinically indicated

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